

Prelab 9.1 Machine Learning 2 - Machine Learning Implementation to Edge Device

Learning Goals

Students will be able to:

1. Install TensorFlow and subsequently load a machine learning (ML) model

1.1 Introduction

In Lab 9, we will implement the machine learning model you developed in Lab 8 to Raspberry Pi. Eventually, you will build up an IIoT smart monitoring system from measuring sensor data in real-time to visualize not only the data but also the result of a machine learning model. This prelab is **1) to install TensorFlow in Raspberry Pi** and **2) to load the model you finally developed in Lab9.**

1.2 TensorFlow Installation on Raspberry Pi

To install TensorFlow, open 'Terminal' on Raspberry Pi and please follow the steps below.

1. To keep Raspberry Pi packages up to date, run the command lines below.



Raspberry Pi - Terminal

```
sudo apt update
```

```
sudo apt upgrade -y
```

2. Install Conda, python 3.11, and tensorflow.

- Conda is a package and environment manager that creates isolated Python environments with specific versions of Python and libraries.



Raspberry Pi - Terminal

```
cd ~
wget -O Miniforge3.sh https://github.com/conda-
forge/miniforge/releases/latest/download/Miniforge3-Linux-
aarch64.sh
```

```
pi@noblinmyeong: ~  
File Edit Tabs Help  
  
pi@noblinmyeong:~$ cd ~  
wget -O Miniforge3.sh https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-Linux-aarch64.sh  
--2026-01-13 18:17:33-- https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-Linux-aarch64.sh  
Resolving github.com (github.com)... 140.82.114.3  
Connecting to github.com (github.com)[140.82.114.3]:443... connected.  
HTTP request sent, awaiting response... 302 Found  
Location: https://github.com/conda-forge/miniforge/releases/download/25.11.0-1/Miniforge3-Linux-aarch64.sh [following]  
--2026-01-13 18:17:33-- https://github.com/conda-forge/miniforge/releases/download/25.11.0-1/Miniforge3-Linux-aarch64.sh  
Reusing existing connection to github.com:443.  
HTTP request sent, awaiting response... 302 Found  
Location: https://release-assets.githubusercontent.com/github-production-release-asset/221584272/a59c9980-2d5a-438c-a777-f00ac98fc3b7?sp=r&sv=2018-11-09&sr=b&sp_r=https%3Ese%2026-01-14T00%3A07%3A21Z&rscsd=attachment%3B+filename%3DMiniforge3-Linux-aarch64.sh&rscst=application%2Foctet-stream&soid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-01-13T23%3A06%3A24Z&ske=2026-01-14T00%3A07%3A21Z&sks=b&skv=2018-11-09&sig=asxHHKTNEzPcj%2BoJA7XrQajXhpSemmgonXCxZsfZoIc%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiIsInR5cCI6IkpzcmMiOiJnaXRodWUiY29tIiwiaXYXNkIjoicmVsZWZfZS1hc3NldHMuz2l0ahVidXNlcmlmVnbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4CiMTc2ODM0ODAlMywibmJiImJoYXNkIjoxNzY4MzQ2MjUzLCJwYXRoIjoicmVsZWZfZWFzc2V0CHJvZHVhdGlvbi5ibG9iLmNmcmUud2luZG93cy5uZXQifQ.MbQ-jes_2EYSMeeyp5b1mFy61gbQoTiL
```

```
bash Miniforge3.sh
```

PRESS Enter, TYPE Yes, TYPE Yes, Yes, Yes...

pi@nabinmyeong: ~

File Edit Tabs Help

```
pi@nabinmyeong:~$ bash Miniforge3.sh

Welcome to Miniforge3 25.11.0-1

In order to continue the installation process, please review the license agreement.
Please, press ENTER to continue
>>>
Miniforge installer code uses BSD-3-Clause license as stated below.

Binary packages that come with it have their own licensing terms
and by installing miniforge you agree to the licensing terms of individual packages as well. They include different OSI-approved licenses including the GNU General Public License and can be found in pkgs/<pkg-name>/info/licenses folders.

Miniforge installer comes with a bootstrapping executable that is used when installing miniforge and is deleted after miniforge is installed. The bootstrapping executable uses micromamba, cllil1, cpp-filessystem, curl, c-ares, krb5, libarchive, libz, lz4, nghttp2, openssl, libsol, libmammajson, reproce and zstd which are licensed under BSD-3-Clause, MIT and OpenSSL licenses. Licenses and copyright notices of these projects can be found at the following URL:
https://github.com/conda-forge/micromamba-feedstock/tree/master/recipe.
```

pi@nabinmyeong: ~

File Edit Tabs Help

```
Linking mamba-2.4.0-h49008fc_1
Linking zstandard-0.25.0-py312hd4f8a7_1
Linking conda-package-streaming-0.12.0-pyhdd8ed1ab_0
Linking urllib3-2.5.0-pyhdd8ed1ab_0
Linking requests-2.32.5-pyhdd8ed1ab_0
Linking conda-package-handling-2.4.0-pyth7909ff3_2
Linking conda-25.11.0-py312h99ef985_0
Linking conda-libmamba-solver-25.11.0-pyhdd8ed1ab_0

Transaction finished

Installation finished.
Do you wish to update your shell profile to automatically initialize conda? This will activate conda on startup and change the command prompt when activated.
If you'd prefer that conda's base environment not be activated on startup, run the following command when conda is activated:

conda config --set auto_activate_base false

Note: You can undo this later by running 'conda init --reverse $SHELL'.

Proceed with initialization? [yes/no]
[no] >>> yes
```

pi@nabinmyeong: ~

File Edit Tabs Help

```
You do accept the license terms? [yes/no]
>>>
Please answer 'yes' or 'no':
>>>
Please answer 'yes' or 'no':
>>>
Please answer 'yes' or 'no':
>>>
Please answer 'yes' or 'no':
>>>
Please answer 'yes' or 'no':
>>>
Please answer 'yes' or 'no':
>>>
Please answer 'yes' or 'no':
>>>
Please answer 'yes' or 'no':
>>>
Please answer 'yes' or 'no':
>>>
Yes

Miniforge3 will now be installed into this location:
```

pi@nabinmyeong: ~

File Edit Tabs Help

```
No change /home/pi/.bashrc
No action taken.

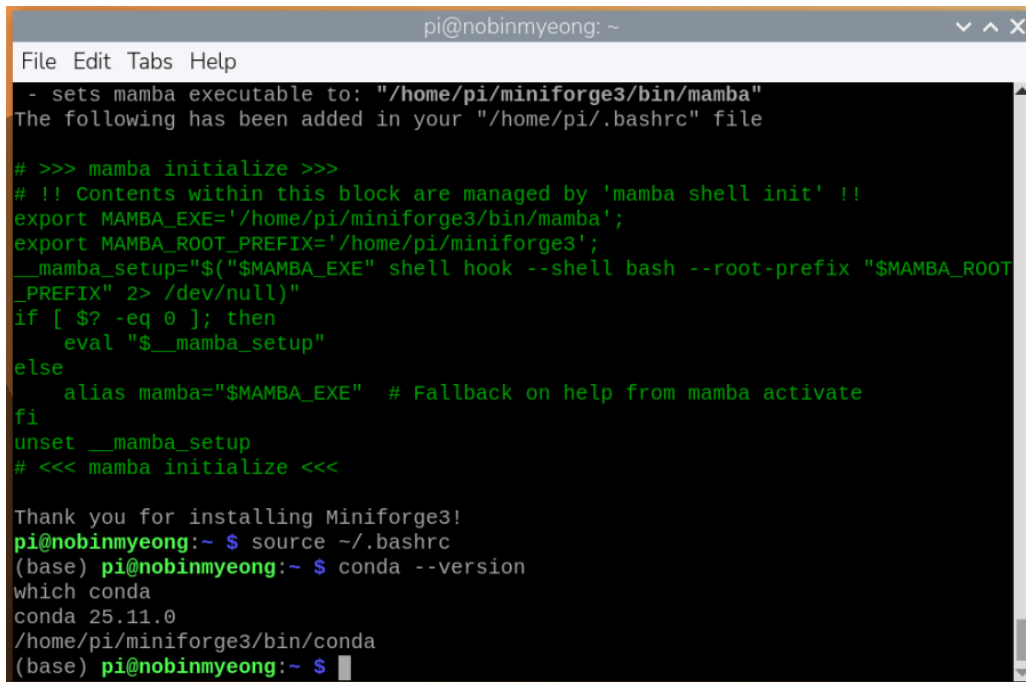
Running 'shell init', which:
- modifies RC file "/home/pi/.bashrc"
- generates config for root prefix: "/home/pi/miniforge3"
- sets mamba executable to: "/home/pi/miniforge3/bin/mamba"
The following has been added in your "/home/pi/.bashrc" file

# >>> mamba initialize >>>
# !! Contents within this block are managed by 'mamba shell init' !!
export MAMBA_EXE="/home/pi/miniforge3/bin/mamba"
export MAMBA_ROOT_PREFIX="/home/pi/miniforge3";
# "MAMBA_SETUP" is the "SMAMBA_EXE" shell hook -> /dev/null"
PREFIX" > /dev/null");
if [ $? -eq 0 ]; then
    eval "$_mamba_setup"
else
    alias mambas="SMAMBA_EXE" # Fallback on help from mamba activate
fi;
unset _mamba_setup
# <<< mamba initialize <<<

Thank you for installing Miniforge3!
pi@nabinmyeong:~$
```

```
source ~/.bashrc
```

```
conda --version
which conda
```



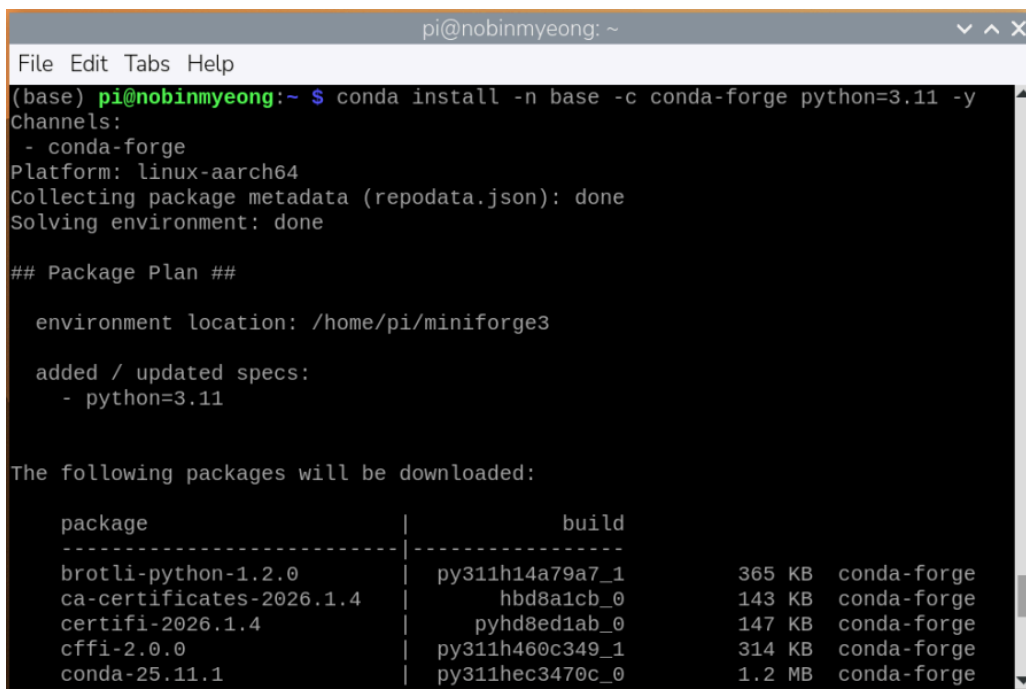
```
pi@nobinmyeong: ~
File Edit Tabs Help
- sets mamba executable to: "/home/pi/miniforge3/bin/mamba"
The following has been added in your "/home/pi/.bashrc" file

# >>> mamba initialize >>>
# !! Contents within this block are managed by 'mamba shell init' !!
export MAMBA_EXE='/home/pi/miniforge3/bin/mamba';
export MAMBA_ROOT_PREFIX='/home/pi/miniforge3';
__mamba_setup="$("$MAMBA_EXE" shell hook --shell bash --root-prefix "$MAMBA_ROOT_PREFIX" 2> /dev/null)"
if [ $? -eq 0 ]; then
    eval "$__mamba_setup"
else
    alias mamba="$MAMBA_EXE" # Fallback on help from mamba activate
fi
unset __mamba_setup
# <<< mamba initialize <<<

Thank you for installing Miniforge3!
pi@nobinmyeong:~ $ source ~/.bashrc
(base) pi@nobinmyeong:~ $ conda --version
which conda
conda 25.11.0
/home/pi/miniforge3/bin/conda
(base) pi@nobinmyeong:~ $
```

```
conda install -n base -c conda-forge python=3.11 -y
```

It usually takes a lot of time... please be patient while the program installs...



```
pi@nobinmyeong: ~
File Edit Tabs Help
(base) pi@nobinmyeong:~ $ conda install -n base -c conda-forge python=3.11 -y
Channels:
 - conda-forge
Platform: linux-aarch64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

environment location: /home/pi/miniforge3

added / updated specs:
 - python=3.11

The following packages will be downloaded:

package | build | size | channel
-----|-----|-----|-----
brotli-python-1.2.0 | py311h14a79a7_1 | 365 KB | conda-forge
ca-certificates-2026.1.4 | hbd8a1cb_0 | 143 KB | conda-forge
certifi-2026.1.4 | pyhd8ed1ab_0 | 147 KB | conda-forge
cffi-2.0.0 | py311h460c349_1 | 314 KB | conda-forge
conda-25.11.1 | py311hec3470c_0 | 1.2 MB | conda-forge
```

```
python -V
```

```
pi@nobinmyeong: ~  
File Edit Tabs Help  
pycosat 0.6.6-py312hcd1a082_3 --> 0.6.6-py311h1935  
2d5_3  
zstandard 0.25.0-py312hd41f8a7_1 --> 0.25.0-py311h51c  
fe5d_1  
  
Downloading and Extracting Packages:  
  
Preparing transaction: done  
Verifying transaction: done  
Executing transaction: done  
(base) pi@nobinmyeong:~ $ python -V  
Python 3.11.14  
(base) pi@nobinmyeong:~ $
```

`python -m pip install --upgrade pip setuptools wheel`

```
pi@nobinmyeong: ~  
File Edit Tabs Help  
pycosat 0.6.6-py312hcd1a082_3 --> 0.6.6-py311h1935  
2d5_3  
zstandard 0.25.0-py312hd41f8a7_1 --> 0.25.0-py311h51c  
fe5d_1  
  
Downloading and Extracting Packages:  
  
Preparing transaction: done  
Verifying transaction: done  
Executing transaction: done  
(base) pi@nobinmyeong:~ $ python -V  
Python 3.11.14  
(base) pi@nobinmyeong:~ $ python -m pip install --upgrade pip setuptools wheel  
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple  
Requirement already satisfied: pip in ./miniforge3/lib/python3.11/site-packages  
(25.3)  
Requirement already satisfied: setuptools in ./miniforge3/lib/python3.11/site-pa  
ckages (80.9.0)  
Requirement already satisfied: wheel in ./miniforge3/lib/python3.11/site-package  
s (0.45.1)  
(base) pi@nobinmyeong:~ $
```

`python -m pip install tensorflow`

It usually takes a lot of time... please be patient while the program installs...

```
pi@nobinmyeong: ~  
File Edit Tabs Help  
(base) pi@nobinmyeong:~ $ python -m pip install tensorflow  
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple  
Collecting tensorflow  
  Using cached tensorflow-2.20.0-cp311-cp311-manylinux_2_17_aarch64.manylinux2014_aarch64.whl.metadata (4.5 kB)  
Collecting absl-py>=1.0.0 (from tensorflow)  
  Using cached https://www.piwheels.org/simple/absl-py/absl_py-2.3.1-py3-none-any.whl (135 kB)  
Collecting astunparse>=1.6.0 (from tensorflow)  
  Using cached https://www.piwheels.org/simple/astunparse/astunparse-1.6.3-py2.py3-none-any.whl (12 kB)  
Collecting flatbuffers>=24.3.25 (from tensorflow)  
  Using cached https://www.piwheels.org/simple/flatbuffers/flatbuffers-20181003210633-py2.py3-none-any.whl (14 kB)  
Collecting gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 (from tensorflow)  
  Using cached https://www.piwheels.org/simple/gast/gast-0.7.0-py3-none-any.whl (22 kB)  
Collecting google_pasta>=0.1.1 (from tensorflow)  
  Using cached https://www.piwheels.org/simple/google-pasta/google_pasta-0.2.0-py3-none-any.whl (57 kB)  
Collecting libclang>=13.0.0 (from tensorflow)  
  Using cached libclang-18.1.1-py2.py3-none-manylinux2014_aarch64.whl.metadata (5.2 kB)  
Collecting opt_einsum>=2.3.2 (from tensorflow)
```

```
python -c "import tensorflow as tf; print(tf.__version__)"
```

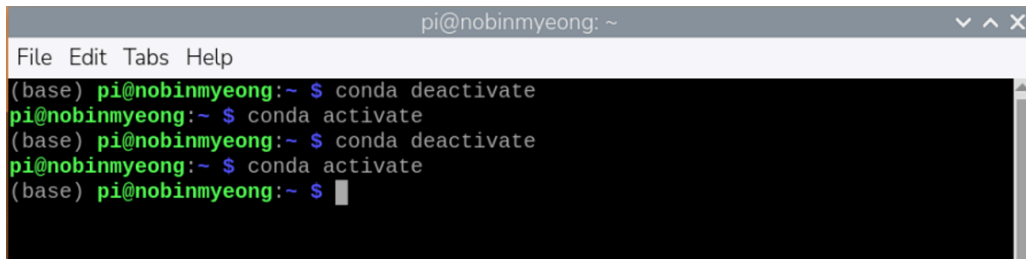
```
pi@nobinmyeong: ~  
File Edit Tabs Help  
Using cached optree-0.18.0-cp311-cp311-manylinux_2_26_aarch64.manylinux_2_28_aarch64.whl (361 kB)  
Using cached pillow-12.1.0-cp311-cp311-manylinux_2_27_aarch64.manylinux_2_28_aarch64.whl (6.3 MB)  
Installing collected packages: namex, libclang, flatbuffers, wrapt, typing_extensions, termcolor, tensorboard-data-server, six, pygments, protobuf, pillow, opt_einsum, numpy, mdurl, markupsafe, markdown, gast, absl-py, werkzeug, optree, ml_dtypes, markdown-it-py, h5py, grpcio, google_pasta, astunparse, tensorboard, rich, keras, tensorflow  
Successfully installed absl-py-2.3.1 astunparse-1.6.3 flatbuffers-20181003210633 gast-0.7.0 google_pasta-0.2.0 grpcio-1.76.0 h5py-3.15.1 keras-3.13.0 libclang-18.1.1 markdown-3.10 markdown-it-py-4.0.0 markupsafe-3.0.3 mdurl-0.1.2 ml_dtypes-0.5.4 namex-0.1.0 numpy-2.4.1 opt_einsum-3.4.0 optree-0.18.0 pillow-12.1.0 protobuf-6.33.4 pygments-2.19.2 rich-14.2.0 six-1.17.0 tensorboard-2.20.0 tensorboard-data-server-0.7.2 tensorflow-2.20.0 termcolor-3.3.0 typing_extensions-4.15.0 werkzeug-3.1.5 wrapt-2.0.1  
(base) pi@nobinmyeong:~ $ python -c "import tensorflow as tf; print(tf.__version__)"  
/home/pi/miniforge3/lib/python3.11/site-packages/keras/src/export/tf2onnx_lib.py:8: FutureWarning: In the future `np.object` will be defined as the corresponding NumPy scalar.  
  if not hasattr(np, "object"):  
2.20.0  
(base) pi@nobinmyeong:~ $
```

3. Done! Reboot Raspberry Pi

 **Raspberry Pi - Terminal**

```
sudo reboot
```

4. Tip! To switch between the Conda environment and the original environment, use the commands shown below.

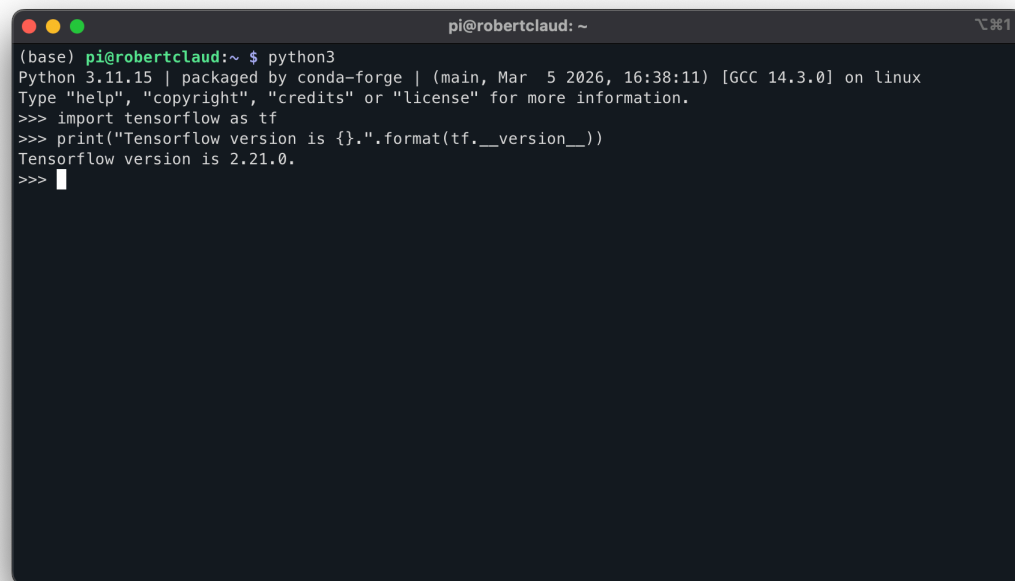
A terminal window titled 'pi@noblinmyeong: ~' with a menu bar (File, Edit, Tabs, Help). The terminal shows a sequence of commands to deactivate and activate a Conda environment. The prompt changes from '(base) pi@noblinmyeong:~' to 'pi@noblinmyeong:~' when the environment is activated.

```
(base) pi@noblinmyeong:~ $ conda deactivate
pi@noblinmyeong:~ $ conda activate
(base) pi@noblinmyeong:~ $ conda deactivate
pi@noblinmyeong:~ $ conda activate
(base) pi@noblinmyeong:~ $
```

Now, you are ready to use TensorFlow on Raspberry Pi! Perform Task 1 to see if TensorFlow is installed successfully on your Raspberry Pi.

Task 1.1

By running the example Python script below, check TensorFlow version as Figure 1 below and then attach it to the report.

A terminal window titled 'pi@robertclaud: ~' with a menu bar (File, Edit, Tabs, Help). The terminal shows a Python script being executed to check the TensorFlow version. The output shows the TensorFlow version is 2.21.0.

```
(base) pi@robertclaud:~ $ python3
Python 3.11.15 | packaged by conda-forge | (main, Mar  5 2026, 16:38:11) [GCC 14.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import tensorflow as tf
>>> print("Tensorflow version is {}".format(tf.__version__))
Tensorflow version is 2.21.0.
>>>
```

Raspberry Pi - Terminal & Python

```
$ python3
>>> import tensorflow as tf
>>> print("TensorFlow version is {}".format(tf.__version__))
```

```
pi@noblinmyeong: ~  
File Edit Tabs Help  
(base) pi@noblinmyeong:~ $ python3  
Python 3.11.14 | packaged by conda-forge | (main, Oct 22 2025, 22:39:18) [GCC 14  
.3.0] on linux  
Type "help", "copyright", "credits" or "license" for more information.  
>>> import tensorflow as tf  
/home/pi/miniforge3/lib/python3.11/site-packages/keras/src/export/tf2onnx_lib.py  
:8: FutureWarning: In the future `np.object` will be defined as the correspondin  
g NumPy scalar.  
if not hasattr(np, "object"):  
>>> print("TensorFlow version is {}".format(tf.__version__))  
TensorFlow version is 2.20.0.  
>>> █
```

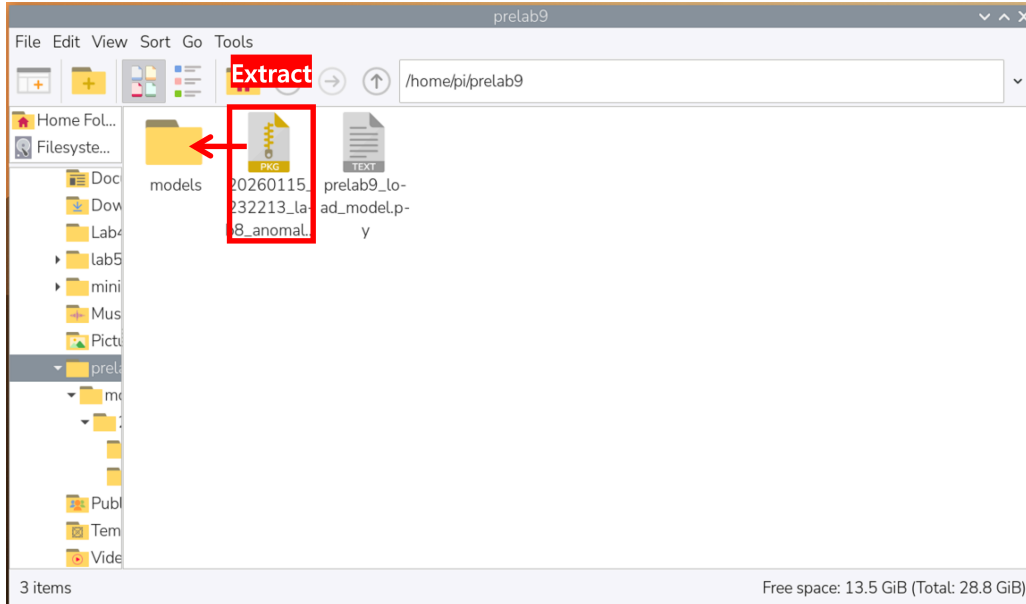
Figure 1 TensorFlow version check, your version should be higher than 2.15.0

1.3 Load Machine Learning Model Using TensorFlow on Raspberry Pi

Let's load the model on Raspberry Pi you finally developed in Lab8. Please follow the steps below.

1. Make a working directory in which your model and Python file will be.

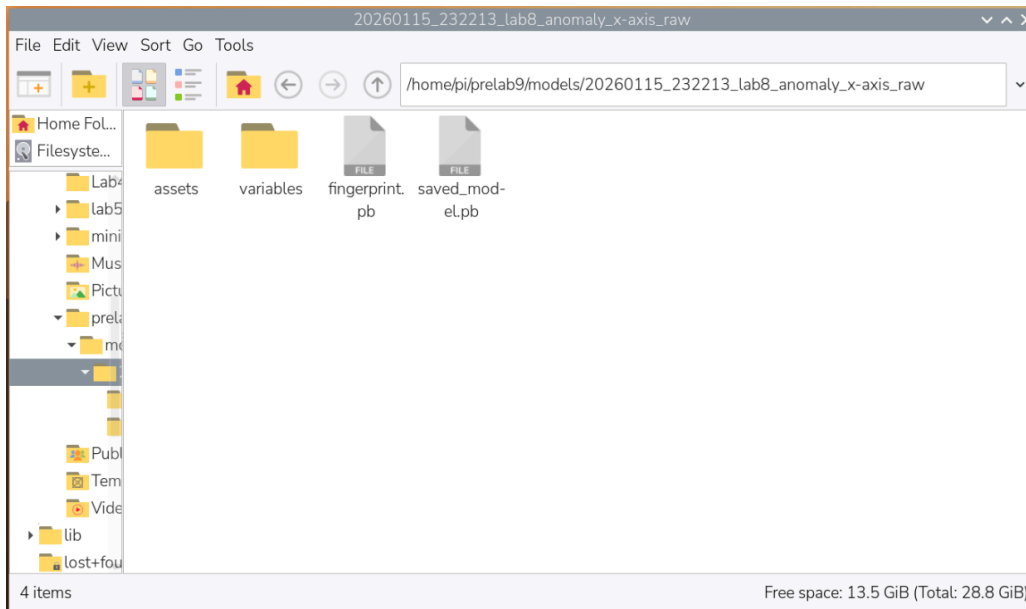
a. The directory example: '/home/pi/prelab9/models'



2. Copy and paste the best model directory you created in Lab 8 to Raspberry Pi directory ('/home/pi/prelab9/models')

a. Check the model folder is moved to the Raspberry Pi

b. In this example, the model folder name is '20260115_222313_lab8_anomaly_x-axis_raw'.



3. Create a Python file as 'prelab9_load_model.py' on the working directory. In this example, the working directory is '/home/pi/prelab9/models/'.
4. Code the Python script below in the created Python file ('prelab9_load_model.py') and then save it.

Python - Python3 (prelab9_load_model.py)

```
import tensorflow as tf

MODEL_DIR =
"/home/pi/prelab9/models/20260115_232213_lab8_anomaly_x-
axis_raw"

loaded = tf.saved_model.load(MODEL_DIR)
infer = loaded.signatures["serving_default"]

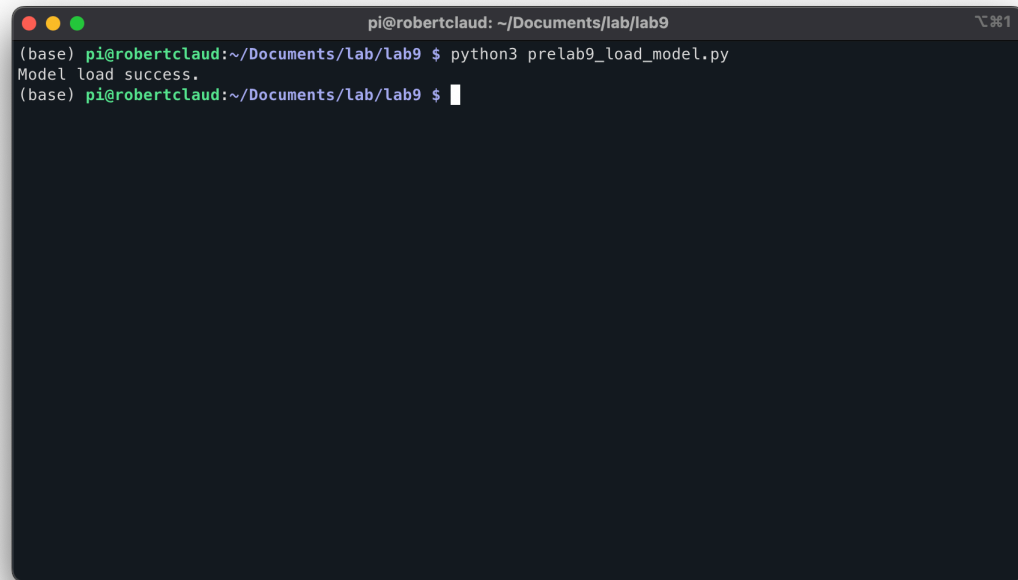
_ = infer(tf.zeros((1, 1000), dtype=tf.float32))

print("Model load success.")
```

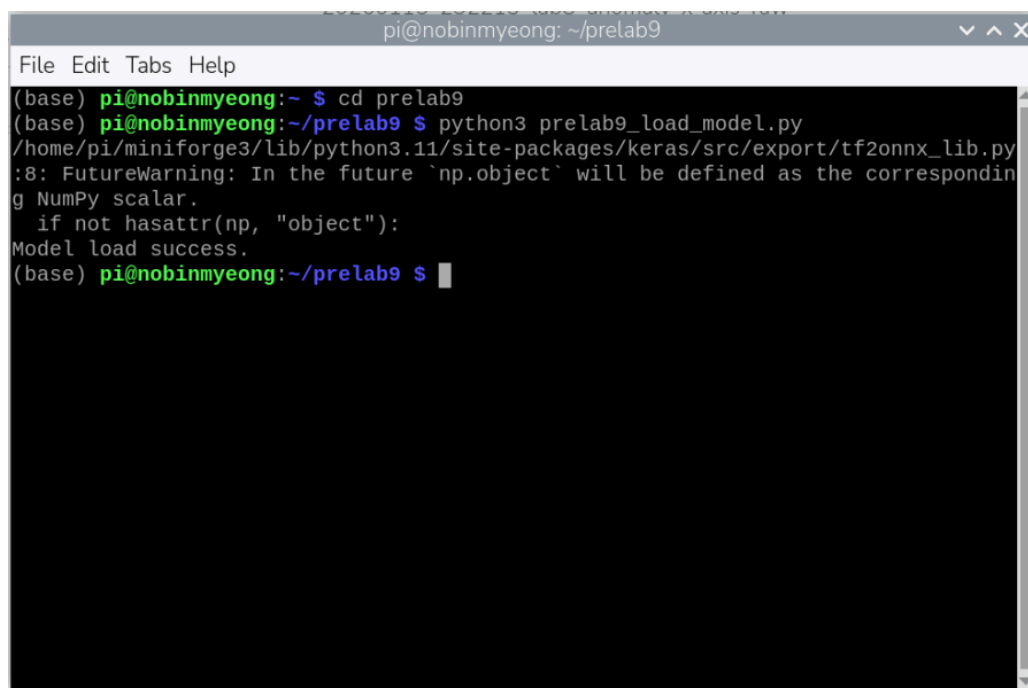
Now, you are ready to use the model you created in Lab9 to predict running conditions of the AFF on Raspberry Pi. Perform Task 1.2 to see if the model is loaded successfully. Moreover, it is highly recommended to review the previous labs to build up the entire monitoring system in Lab9.

Task 1.2

1. After performing the steps (load the model) above, run the Python file. Is the result as you expected?
 2. Capture the result of Terminal or Shell of Thonny as Figure 2 and attach it to the report.
-



```
pi@robertclaud: ~/Documents/lab/lab9
(base) pi@robertclaud:~/Documents/lab/lab9 $ python3 prelab9_load_model.py
Model load success.
(base) pi@robertclaud:~/Documents/lab/lab9 $
```



```
pi@noblinmyeong: ~/prelab9
File Edit Tabs Help
(base) pi@noblinmyeong:~ $ cd prelab9
(base) pi@noblinmyeong:~/prelab9 $ python3 prelab9_load_model.py
/home/pi/miniforge3/lib/python3.11/site-packages/keras/src/export/tf2onnx_lib.py:8: FutureWarning: In the future `np.object` will be defined as the corresponding NumPy scalar.
  if not hasattr(np, "object"):
Model load success.
(base) pi@noblinmyeong:~/prelab9 $
```

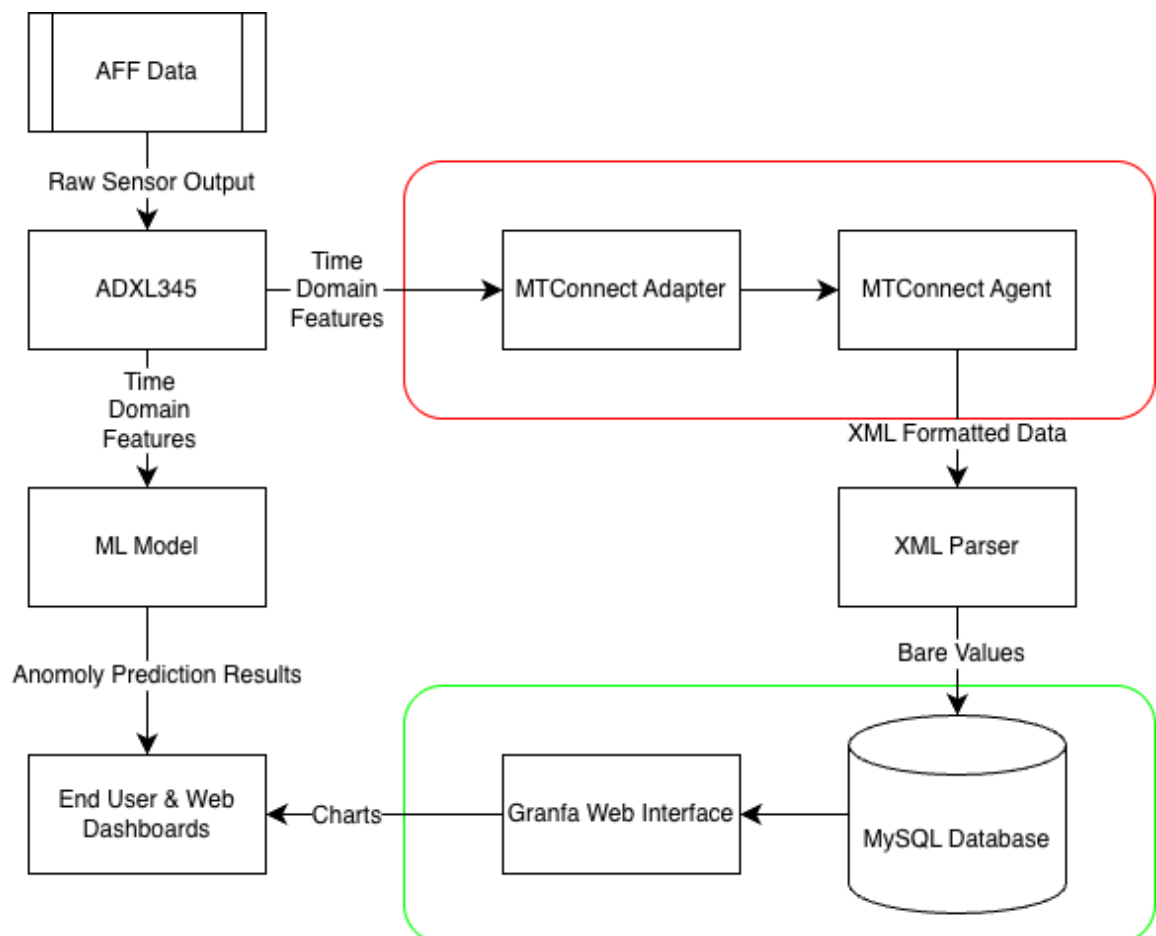
Figure 2 Model load

(Optional) Play with TensorFlow and the Loaded Model

By using the example Python script to load the model and the sample Python scripts in the previous labs, try to run the model by measuring the ADXL345 sensor. In your home setup without attaching the sensor to the AFF, what is the output from the model? Play with the sensor and the model!

Task 1.3 Using the list below, describe the data flow pipeline (system configuration) and software you need to use in order to realize real-time running condition prediction of the AFF.

- Sensor: ADXL345
- Middleware: MTConnect
- Database: MySQL
- Visualization: Grafana
- Monitoring target(s):
 - Time domain features of each axis of the sensor
 - Anomaly prediction results of the AFF



The data flow pipeline for this effort is as follows. First, AFF Data is interpreted by the ADXL345 and parsed by the Raspberry Pi. This allows the time domain features to be extracted and set to both the MTConnect Adapter and a trained ML model. As in lab 7, the pipeline of data goes from the MTConnect Adapter, to the agent which then can get parsed and dumped into the MySQL Database. The raw data can then be viewed on a dashboard created on the Granfa Web Interface. For the ML Model interpretation, the ML model can interpret time domain features and use this to predict anomalous operating conditions. Using this prediction it can then send its own charts to the end user and web dashboards.

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